

Numbers, Sets and Functions

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October 4, 2026

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1 Week 1

1.1 Definitions

Definition 1. *Definition 1.1. A proof is a formal logical argument justifying that a mathematical statement is true, by proceeding from given assumptions to the desired conclusion.*

Definition 2. *Theorem 1.2. The sum of two even numbers is even.*

Definition 3. *A statement represents an expression that: • is a complete sentence by itself, and • is either true or false.*

1.2 Notes

Formal logic can be viewed in two ways. Boolean logic focuses on whether statements are true or false, which is useful in computing. Deductive logic focuses on how conclusions follow from assumptions, as in mathematical proofs.

Let P , Q , and R be statements. The symbols $\wedge$, $\vee$, and $\neg$ mean “and,” “or,” and “not.” The symbol $\Rightarrow$ means “implies,” and $\Leftrightarrow$ means “is logically equivalent to.”

The following laws give logically equivalent ways to write statements:

Commutative laws: $P \wedge Q \Leftrightarrow Q \wedge P$,

$P \vee Q \Leftrightarrow Q \vee P$,

Associative laws: $P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$,

$P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$,

De Morgan’s laws: $\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$,

$\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$,

Distributive laws: $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$,

$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$.

An equivalence can be expressed as implications in both directions:

$P \Leftrightarrow Q$ means $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$.

An implication has the same logical meaning as its contrapositive:

$P \Rightarrow Q \Leftrightarrow (\neg Q) \Rightarrow (\neg P)$.

2 Week 2

2.1 Definitions

2.2 Notes

3 Week 3

3.1 Definitions

3.2 Notes

4 Week 4

4.1 Definitions

4.2 Notes

5 Week 5

5.1 Definitions

5.2 Notes

6 Week 6

6.1 Definitions

6.2 Notes

7 Week 7

7.1 Definitions

7.2 Notes

8 Week 8

8.1 Definitions

8.2 Notes

9 Week 9

9.1 Definitions

9.2 Notes

10 Week 10

10.1 Definitions

10.2 Notes

11 Week 11

11.1 Definitions

11.2 Notes

12 Week 12

12.1 Definitions

12.2 Notes